

12.580

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Let $\mathbf{V}$ be the vector space of all 3×3 matrices with complex entries over the real field.
If

$$\mathbf{W}_1 = \{\mathbf{A} \in \mathbf{V} : \mathbf{A} = \overline{\mathbf{A}}^\top\} \quad \text{and} \quad \mathbf{W}_2 = \{\mathbf{A} \in \mathbf{V} : \text{trace}(\mathbf{A}) = 0\},$$

then find $\dim(\mathbf{W}_1 + \mathbf{W}_2)$.

Solution:

$$\mathbf{V} = \text{all } 3 \times 3 \text{ complex matrices over } \mathbb{R} \quad (1)$$

For one complex element 2 real parameters are required.

$$\dim_{\mathbb{R}}(\mathbf{V}) = 3 \times 3 \times 2 = 18 \quad (2)$$

$$\mathbf{W}_1 = \{\mathbf{A} \in \mathbf{V} : \mathbf{A} = \overline{\mathbf{A}}^\top\} \quad (3)$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ \overline{a_{12}} & a_{22} & a_{23} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} \end{pmatrix} \quad a_{ii} \in \mathbb{R}, \quad a_{ij} \in \mathbb{C}, \quad i < j \quad (4)$$

$$\dim_{\mathbb{R}}(\mathbf{W}_1) = 3 + 3 \times 2 = 9 \quad (5)$$

$$\mathbf{W}_2 = \{\mathbf{A} \in \mathbf{V} : \text{trace}(\mathbf{A}) = 0\} \quad (6)$$

So one element along diagonal has constraint.

$$\dim_{\mathbb{R}}(\mathbf{W}_2) = ((3 \times 3) - 1) \times 2 = 16 \quad (7)$$

$$\dim_{\mathbb{R}}(\mathbf{W}_1 \cap \mathbf{W}_2) = 9 - 1 = 8 \quad (8)$$

$$\dim(\mathbf{W}_1 + \mathbf{W}_2) = \dim(\mathbf{W}_1) + \dim(\mathbf{W}_2) - \dim(\mathbf{W}_1 \cap \mathbf{W}_2) \quad (9)$$

$$= 9 + 16 - 8 \quad (10)$$

$$= 17 \quad (11)$$